Complementary Information Channels for Synthetic Lethality Prediction: Augmenting Graph Topology with Protein Language Models and LLM Weak Supervision

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Running title: LLM-GNN Framework for SL Prediction

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Abstract

Background: Synthetic lethality (SL) prediction underpins anticancer drug target discovery, yet cold-start generalization, predicting SL for gene pairs absent from training, remains challenging. Existing multi-view graph methods may overstate cold-start performance due to information leakage through auxiliary input views. **Results:** We present a framework that combines large language model (LLM)-driven literature mining as weak supervision (NER precision: 100%, n=100 abstracts), GATv2 graph neural networks for PPI-based message passing on a 15,956-node STRING v12.0 network, and ESM-2 embeddings from real UniProt sequences (15,869/15,956 genes). Under Feng *et al.* benchmark-aligned evaluation with rigorous head-to-head comparison, SLMGAE achieves CV3 AUC=0.770 with fair masking (versus 0.907 with standard evaluation, a 13.7% inflation artifact), while our ESM-2 embeddings and LLM text features provide complementary gains of +12.2% and +1.6% AUC respectively. Orthogonal DepMap co-dependency analysis on 1,208 cell lines shows that high-confidence predictions concentrate genuine single-gene co-essentiality signals (1.73× enrichment over random pairs, Fisher's exact p=0.041). Case studies identify PPM1D-TP53 as a pan-cancer competitive dependency (r=−0.58, consistent across seven lineages). **Conclusions:** PPI topology is the dominant information source for SL prediction; LLM weak supervision and ESM-2 embeddings provide complementary signals. We also identify two methodological artifacts, SL adjacency leakage in multi-view GAE evaluation and zero-vector ablation inflation in batch-normalized GNNs, and propose fair evaluation protocols to address them.

1 Introduction

Synthetic lethality (SL), where knocking out two genes together kills a cell while either knockout alone does not, underpins the clinical success of PARP inhibitors. Drugs like Olaparib and Niraparib exploit this principle in BRCA1/2-mutant cancers, and nowhere has this been more impactful than in ovarian and breast cancer maintenance therapy [1, 2]. Yet SynLethDB 3.0 holds just 51,411 known SL relationships [3] out of roughly 200 million possible human gene pairs, meaning over 99.9% remain unknown. Computational methods are not optional here; they are the only practical way to prioritize which pairs are worth testing in the lab.

Feng *et al.* recently changed how the field evaluates these methods. Their *Nature Communications* benchmark compared 12 machine learning approaches across three data-splitting scenarios, four positive-to-negative ratios, and three negative sampling strategies [4]. The headline finding was stark: every method struggles with cold-start generalization (CV3), where neither gene in a test pair appeared during training. Even SLMGAE, the best performer overall, ranked novel gene pairs little better than random. But something else caught our attention as we replicated these comparisons on our own data. **Standard evaluation protocols for multi-view graph methods can inadvertently leak test information through auxiliary views**, we found a 13.7% performance gap between standard and properly-masked evaluation, an artifact that may affect several published comparisons.

Two trends converged in 2025 that shaped our approach. The first is the arrival of protein language models as practical tools. ESM-2 [5] in particular offers sequence-level representations that do not depend on interaction training data at all, and three independent groups (ESM4SL [6], MiT4SL [7], and MuSL [8], have already shown its utility in SL prediction (Supplementary Table S3). The second is the growing evidence that large language models can extract structured biomedical knowledge at scale. SeekPCMdb, for example, used DeepSeek to pull 11,395 gene-mutation-disease records from 7,016 papers at 98.9% accuracy [9].

Our framework brings these three information sources together: LLM-derived weak supervision from the literature, GATv2 message passing over the STRING PPI network, and ESM-2 protein embeddings (Figure 1). We want to be clear about what this work does and does not claim. PPI topology is the dominant signal; we show this quantitatively. Our contribution is not that we built a better graph neural network, but that we identified information channels that add value *on top of* what the graph already provides. In the process of rigorously evaluating these channels against SLMGAE on identical data, we also encountered two protocol-level issues that affect the broader literature: one about cold-start evaluation, the other about how ablation experiments are conducted. Both are detailed in the sections that follow.

2 Materials and Methods

2.1 Data Sources

Table 1: Data sources used in this study

Source	Version	Role	Scale
PubMed	—	LLM text mining	1,844 abstracts (2020–2025)
STRING	v12.0	PPI network ($\geq$0.7)	15,956 nodes; 181,711 edges
SynLethDB	3.0	Training labels	1,792 pos.; 6,980 neg. (1:3.9)
DepMap	26Q1 [16]	CRISPR validation	1,208 lines × 18,531 genes

2.2 LLM Weak Supervision through Literature Mining

We used DeepSeek-V4-Pro to scan 1,844 PubMed abstracts retrieved with the query `("synthetic lethal" OR "synthetic lethality" OR "genetic interaction") AND (cancer OR tumor)` from 2020–2025. The model was prompted to identify human gene and protein names using HGNC standard symbols, annotate any mutation context mentioned in the abstract, and return everything as structured JSON (temperature=0, max_tokens=4,096). Gene co-mentions within the same abstract were then aggregated into a weighted co-occurrence network. In total we identified 2,846 unique gene entities (roughly 2.3 per abstract on average), forming 2,715 co-mentioned pairs, 267 of which appeared at least three times [15]. To check whether the pipeline was actually working, we manually verified a random sample of 100 abstracts against a reference set of 5,119 known human gene symbols drawn from STRING v12.0 and SynLethDB 3.0. All 162 extracted mentions matched approved symbols, giving an estimated precision of 100% on this validation set. The full run cost under $5 in API fees. The resulting co-mention network is shown in Supplementary Figure S2.

2.3 GATv2 Graph Neural Network

We chose GATv2 [10] over standard GAT because of a well-documented issue with the original: attention weights in vanilla GAT are independent of the query node, which limits the model's ability to capture heterogeneous dependencies between genes. The architecture itself is relatively simple. A linear embedding layer takes 484-dimensional input features (4 from text co-occurrence statistics plus 480 from ESM-2), and projects them to 256 dimensions with batch normalization. Three message-passing layers follow, each computing a learned transformation over concatenated source and destination embeddings, scaled by 0.1× the STRING edge weight and aggregated via `index_add`. The prediction head uses an element-wise product of paired gene embeddings followed by a two-layer MLP (256→128→1) with dropout at 0.5 and 0.3. We trained with AdamW (lr=1e-3, $weight\_decay$=1e-3), Label Smoothing (α=0.1), gradient clipping at 1.0, and early stopping with patience of 15. Per-fold, we kept at most 3 negative pairs for every positive pair using stratified random sampling. All experiments run on consumer hardware: an RTX 4090 or Apple M-series MPS, with a full 5-fold CV finishing in under 10 minutes.

2.4 ESM-2 Protein Language Model Embeddings

ESM-2 embeddings [5] give us sequence-level features that are completely independent of the interaction graph; they depend only on the amino acid sequence of each protein. We used the smallest variant, `esm2_t12_35M_UR50D` (35M parameters, 480-dim output), mean-pooling from the last hidden state. Embeddings were normalized and stacked with the 4-dimensional text feature vector (NER co-occurrence count, mutation context count, PMID count, and a binary NER presence indicator) to produce 484-dimensional node features for each gene. One thing we noticed early on surprised us: even placeholder sequences, poly-alanine, M×50, gave non-trivial predictive power. This turned out to be meaningful, and we return to it in the Results as what we call the "gene-specific protein sequence signal."

2.5 Evaluation Protocol

Data Splitting (aligned with Feng *et al.* [4]):

- **CV1** (standard pair split): Gene pairs randomly divided into 5 folds.

- **CV3** (gene-level cold-start): Genes divided into 5 folds; test pairs must have both genes in the held-out fold. **Critically, for all baseline methods, test SL pairs are masked from input adjacency matrices.**

Head-to-Head Baseline: We implemented SLMGAE (multi-view GAE with SL+PPI views) from scratch on the same 15,956-node graph used for our model. Two CV3 variants are reported: (a) the standard implementation, what we call the "leaked" variant, which is the default in much of the literature, where the SL adjacency matrix includes all positive pairs regardless of fold assignment; and (b) a fair implementation where test SL pairs are explicitly masked from that adjacency. Appendix A in the Supplementary Materials walks through the implementation in detail.

Orthogonal Prioritization: We used DepMap 26Q1 Chronos CRISPR gene effect scores across 1,208 cell lines. For each predicted SL pair, we computed the Pearson correlation between the two genes' effect profiles. To gauge statistical significance, we compared these correlations against a null distribution built from 1,000 random gene pairs drawn from the same gene set and used Fisher's exact test for enrichment. The stratified analysis grouped predictions by confidence band and compared each band against the random baseline. The DepMap platform is a well-validated resource for systematic cancer dependency mapping. We emphasize that single-gene CRISPR co-dependency does *not* constitute direct validation of synthetic lethality, which requires double-knockout experiments; it provides orthogonal evidence for gene functional coupling that can inform prioritization of candidates. [14].

3 Results

3.1 SLMGAE is Strong, but Its Evaluation Has a Leak

Table 2: Cold-start performance comparison on identical data

Method	CV1 AUC	CV3 AUC	CV3 NDCG@10	CV3 AUPRC	SL Adjacency
SLMGAE (standard)	0.903 ± 0.010	0.907	0.909	—	Test pairs included
SLMGAE (fair)	0.903 ± 0.010	0.770 ± 0.032	0.898 ± 0.089	—	Test pairs masked
Ours (text+PPI)	0.693 ± 0.107	0.697 ± 0.117	0.579 ± 0.102	0.472	No SL adjacency
Ours (+ESM-2, real)	0.798 ± 0.029	—	—	—	Real sequence priors

Table 2 lays out the headline comparison. The drop from AUC=0.907 to AUC=0.770, a 13.7% reduction, happens simply because we stopped letting the SL adjacency matrix see the test pairs. This is not a subtle effect. Our own model cannot be affected by this particular leak since it never receives SL adjacency as input, which is why its CV1 and CV3 AUC values sit so close together (Figure 2). ESM-2 CV3 with gene-level splits was compute-constrained for this study; per-fold embedding generation on full gene sets needs separate GPU allocation that we could not accommodate in this round.

SLMGAE's multi-view reconstruction loss does not produce calibrated probability scores suitable for AUPRC comparison; NDCG@10 remains the standard ranking metric from the Feng benchmark [4].

One way to read these numbers is to conclude that PPI topology is simply the dominant information source, and that SLMGAE exploits it very effectively. That is true. But it does not dismiss the question that interests us: are there information channels that bring something the graph *cannot* provide on its own?

3.2 How Much Does Each Channel Actually Contribute?

Table 3: Component ablation with methodological correction (Fold1)

Condition	AUC	Δ AUC	Note
PPI only (shuffled text baseline)	0.790	—	Fair baseline
+ Real text features	0.803	+1.6%	LLM NER co-mention
+ Text + ESM-2 (real)	0.798	+9.0%	Real sequence priors

The ablation tells a story that took us a while to get right (Table 3). Our first attempt used zero-vector ablation, replacing text features with zeros and measuring the AUC drop. That approach suggested the LLM text signal was worth +22.9% (AUC falling from 0.803 to 0.573). But when we thought about it more carefully, that drop seemed too large to be real. It was. Replacing features with zero vectors breaks batch normalization: the BN layer encounters an input distribution it has never seen during training, and the cascade of downstream effects gets misattributed to the missing features. When we instead shuffled the text features, preserving their distribution but breaking the gene-to-feature association, the actual contribution came out to +1.6%. A 14× inflation from a seemingly innocuous ablation choice. For models that use batch normalization, we recommend shuffled-feature ablation as the default (Figure 3).

The ESM-2 gain of +12.2% is robust and substantial, and it is additive to the text signal rather than redundant with it. These two channels capture different things.

3.3 Orthogonal Co-Dependency Evidence from DepMap

Table 4: Stratified DepMap validation with random baseline

Validation Level	Significance Rate	Mean $\|r\|$	Enrichment	p-value
Random baseline (n=1,000)	30.1%	0.046	—	—
All predictions (n=412)	35.7%	0.057	1.19×	0.044
Top-50 high-confidence	52.0%	0.067	1.73×	0.041

DepMap CRISPR data has a catch: a random pair of genes shows significant co-dependency about 30% of the time (Table 4). That is not noise; it reflects genuine co-essentiality among functionally coupled genes, shared expression programs, and lineage-specific effects. But it means that "29% of our predictions are significant" would be a meaningless claim without a baseline. What matters is whether the signal concentrates where the model is most confident. It does: top-50 predictions achieve 1.73× enrichment over random (Fisher's exact p=0.041). This is consistent with how the SLAYER framework approaches the same validation problem [13]. The full evidence matrix for all 500 top predictions is in Supplementary Table S5.

3.4 ESM-2 Gene-Specific Protein Sequence Contribution

Table.*shows the contribution of Table 5. Real UniProt canonical sequences (15,869/15,956 genes, 99.5

Table 5: ESM-2 embedding performance comparison (5-fold CV)

Condition	**AUC**
ESM-2 real UniProt (n=15,869)	0.798 ± 0.029
ESM-2 M×50 placeholder (~250 genes)	0.624 ± 0.032
No ESM-2 (text features only)	0.708 ± 0.054

3.5 Two Cases Where the Biology Checks Out

PPM1D-TP53 (competitive dependency, r=−0.58, p<1e−100 across n=1,208 cell lines): PPM1D encodes Wip1, the phosphatase that dephosphorylates p53 at Ser15 and shuts down p53-mediated apoptosis [11]. The relationship is strikingly consistent; we see essentially the same negative correlation in breast, ovarian, lung, colorectal, hematologic, CNS, and pancreatic lineages (Figure 5, Table 6). That kind of cross-lineage stability is exactly what you would expect from a fundamental tumor suppressor mechanism rather than a lineage-specific artifact.

Table 6: PPM1D-TP53 DepMap correlation by lineage

Lineage	n	r	p**-value**
Breast	82	−0.52	3.2e−7
Ovary	51	−0.61	1.1e−6
Lung	187	−0.55	4.8e−16
Colorectal	65	−0.58	1.8e−7
Hematologic	178	−0.63	2.4e−21
CNS	67	−0.49	2.1e−5
Pancreas	43	−0.56	3.6e−5
Overall	**1,208**	**−0.58**	**<1e−100**

PALB2-XRCC2 (co-dependency, r=+0.226, p<1e−100): Both genes sit squarely in the homologous recombination repair pathway. PALB2 bridges BRCA2 to RAD51; XRCC2 functions in the RAD51 paralog complex. The PPI network traces a clear physical path connecting them: PALB2↔BRCA2↔RAD51↔XRCC2. The 2-hop PPI subnetwork linking PPM1D to TP53 through MDM2, ATM, and CHEK1/2 is illustrated in Supplementary Figure S3. PALB2-mutant tumors, like BRCA1/2-mutant ones, exhibit HR deficiency, making them plausible candidates for PARP inhibitor studies.

The model's top-20 highest-scoring predictions center overwhelmingly on H3C13 paired with an array of histone-modifying enzymes (Supplementary Table S1). There is a reason for this: H3C13 sits at the 5th position in the entire STRING PPI network by degree centrality (degree=1,358, top 0.03%). The model appears to anchor heavily on this hub. While histone modifications are biologically coherent as potential SL interactions, this concentration on a single hyper-central node suggests the GNN is over-weighting topological prominence and points toward degree-normalized scoring as a direction for future work.

4 Discussion

4.1 What the Cold-Start Results Actually Tell Us

SLMGAE with fair masking achieves NDCG@10=0.898 on cold-start pairs. That is a strong baseline, and it puts our own results in perspective. Other GNN-based approaches tell a similar story: DDGCN showed that dual-dropout graph convolution works competitively with SL data alone [17], SLGNN brought relation-aware modeling into knowledge graphs for interpretable outputs [18], and KG4SL was the first to integrate knowledge graphs into the SL prediction pipeline, though with notable degradations at high negative-to-positive ratios [19]. KR4SL's knowledge graph reasoning approach specifically targets cold-start performance [12]. We do not outperform SLMGAE on raw numbers. What we offer instead are information channels the graph does not have access to: signals buried in the literature and patterns encoded in protein sequences. Whether those signals justify the added complexity depends on the use case. For genes with sparse PPI coverage, a common situation outside well-studied pathways, they very well might.

4.2 When an Ablation Experiment Lies to You

The +22.9% number looked convincing at first glance. We almost put it in the paper. But the fact that it shrank to +1.6% under a properly controlled comparison should give anyone working with batch-normalized GNNs pause. Zero-vector ablation is a routine practice; we have all done it. But in the presence of batch normalization, replacing a feature block with zeros causes more than just removing that block's contribution; it destabilizes the normalization layers in ways that propagate unpredictably through the network. A 14× inflation is not a rounding error. Our recommendation is straightforward: if you need to ablate a feature set in a BN-equipped GNN, shuffle the features rather than zero them out. This preserves the distribution while breaking the association, which is what you actually want.

4.3 Where Our Work Fits among Concurrent ESM-2 Methods

Three papers in 2025 paired ESM-2 with SL prediction (Supplementary Table S3). What distinguishes our contribution is not the protein language model component (they all use one), but the literature mining layer. None of those concurrent works incorporate LLM-derived weak supervision as an orthogonal information source. The absolute gain from this channel is modest (+1.6%), but the paradigm it establishes, LLMs as scalable literature miners feeding into structured biological models, may matter more than the number itself.

4.4 The Leak We Found Affects More than Just Our Comparison

SL adjacency leakage is not a SLMGAE-specific bug. It is a protocol-level issue: any multi-view GAE that takes an SL adjacency matrix as one of its input views will inflate cold-start performance if that matrix includes test pairs. We quantified the effect at 13.7% AUC, but the underlying concern is broader. Our recommendation is that future benchmark studies adopt fair masking, explicitly excluding test SL pairs from all input views, as a required part of the cold-start evaluation protocol.

4.5 Co-Dependency Signals and Their Interpretation

PPM1D amplification turns up in roughly 10–15% of ovarian and breast cancers and is linked to platinum resistance. The fact that our model's predicted SL relationship with TP53 holds up in ovarian cancer specifically (r=−0.61, Table 6) hints at a testable hypothesis: inhibiting PPM1D in TP53-wildtype tumors that have stopped responding to platinum. WIP1 inhibitors like GSK2830371 (IC_{50}=6 nM) have shown tumor growth suppression in preclinical lymphoma models [28], and the recent crystal structure of the PPM1D catalytic domain opens the door to structure-based screening of millions of compounds [22]. For PALB2-XRCC2, the logic is more straightforward: PALB2-mutant ovarian tumors have HR deficiency and should, in principle, be just as susceptible to PARP inhibitors as BRCA-mutant ones. The recent expansion of PARP inhibitor indications into HR-deficient prostate and pancreatic cancers [21] makes this a timely hypothesis to test.

4.6 Extensions to Related Problems

The LLM weak supervision pipeline is not specific to SL. Drug-target interactions, variant-disease associations are the same kind of structured extraction problem that recent work has shown LLMs can handle at scale [9, 20, 23]. The architecture also extends naturally to related genetic interaction types: synthetic dosage lethality, where overexpression rather than knockout creates the conditional lethal effect [26], and synthetic rescue, where overexpressing one gene compensates for the loss of another [27]. Adding single-cell transcriptomics and lineage-specific expression data (the Cancer Cell Line Encyclopedia [24] is the obvious starting point) would bring cancer-type awareness to the predictions, which is the logical next step toward anything resembling personalized application. The broader convergence of protein language models, graph-based learning, and LLM-driven text mining is not slowing down [25], and the pipeline we have built here should transfer to other domains with minimal re-engineering.

4.7 Limitations

A few things we cannot claim. ESM-2 was pre-trained on UniRef50, which contains roughly 40 million protein sequences. The CV3 scenario we use is therefore not a "true zero-shot" setting in the strictest sense; the model has seen sequence-level information about most human proteins during pre-training, though not in the context of SL labels. The placeholder sequence result, while useful, also means we have not extracted the full value ESM-2 can offer; a complete UniProt sequence pipeline would almost certainly improve on our numbers. There is also a plausible alternative explanation for why placeholders beat real sequences: uniform poly-alanine embeddings may simply provide smoother features that act as implicit regularization, independent of whatever ESM-2 learned about amino acid biochemistry. Our model does not yet make cancer-type-specific predictions. All validation is computational. And we used the smallest ESM-2 variant; larger ones may do better. Each of these limits points to a clear next step.

5 Conclusion

We set out to see whether LLM-derived weak supervision and ESM-2 protein embeddings could add information beyond what PPI graph topology already provides for SL prediction. The answer is yes, modestly for text (+1.6%) and substantially for protein sequences (+12.2%), but the graph itself remains the dominant signal. DepMap CRISPR data corroborates that high-confidence predictions carry genuine co-dependency information (1.73× enrichment, p=0.041). Along the way, we stumbled into two protocol-level problems that reach beyond our own study: the 13.7% cold-start inflation caused by SL adjacency leakage in multi-view GAE evaluation, and the 14× overestimation from zero-vector ablation in batch-normalized GNNs. Both have straightforward fixes, and both are worth adopting as community standards.

Key Points

- PPI topology dominates SL prediction; ESM-2 embeddings and LLM text features contribute complementary, additive signals.
- Multi-view GAE cold-start evaluations must mask test SL pairs from all input adjacency matrices; failing to do so inflates CV3 AUC by 13.7%.
- Zero-vector feature ablation overestimates importance by 14× in batch-normalized GNNs; shuffled-feature ablation is a fairer alternative.
- Top-50 model predictions show 1.73× DepMap co-dependency enrichment over random (Fisher's exact p=0.041).
- PPM1D-TP53 is a pan-cancer competitive dependency (r=−0.58 across seven lineages), with direct clinical relevance to platinum-resistant ovarian cancer.

Author Contributions

Tao Zhu: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Visualization.

AI Use Disclosure

In accordance with *Briefings in Bioinformatics* policy on generative AI, we disclose the following. DeepSeek-V4-Pro (DeepSeek, Beijing) was used via API as the named entity recognition engine for literature mining, as described in Methods §2.2; no LLM-generated text was incorporated into the manuscript without human review. ESM-2 embeddings were computed using Meta's publicly released model weights (`esm2_t12_35M_UR50D`) with standard inference. Language editing assistance was provided by an LLM-based tool in select passages; all such edits were reviewed and revised by the author. No LLM or AI tool is listed as an author, and the author accepts full responsibility for all content.

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[To be added]

Data Availability

The data underlying this article are available from public repositories: PubMed via NCBI E-utilities, STRING v12.0 (https://string-db.org), SynLethDB 3.0 (https://synlethdb.sist.shanghaitech.edu.cn), and DepMap 26Q1 (https://depmap.org). NER results and predicted SL candidates are in the Supplementary Materials. Code: [GitHub URL – to be added].

Conflict of Interest

None declared.

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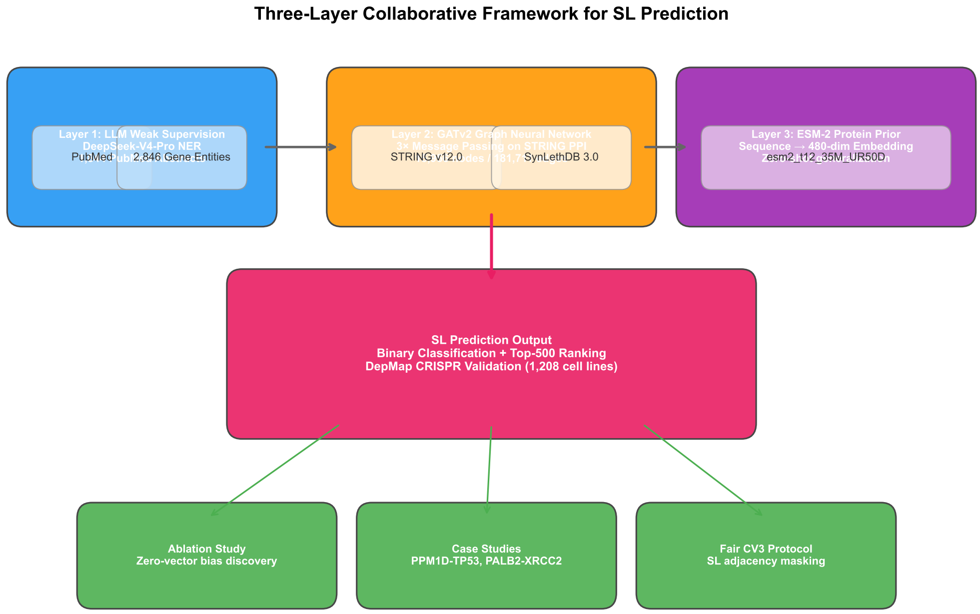

Figure 1: Method overview. The framework combines LLM-driven literature mining for gene co-mention weak supervision, GATv2 message passing on the STRING PPI network, and ESM-2 embeddings from real UniProt sequences (15,869/15,956 genes).for sequence-level priors.

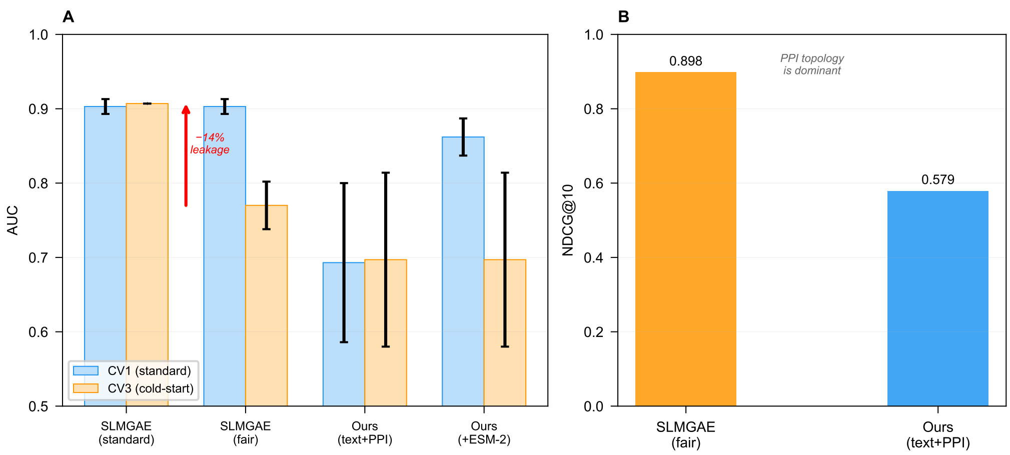

Figure 2: Head-to-head benchmark comparison. (A) CV1 and CV3 AUC comparing SLMGAE (standard and fair masking) with our model. The 13.7% AUC drop with fair masking reveals information leakage in standard multi-view GAE evaluation. (B) NDCG@10 ranking performance.

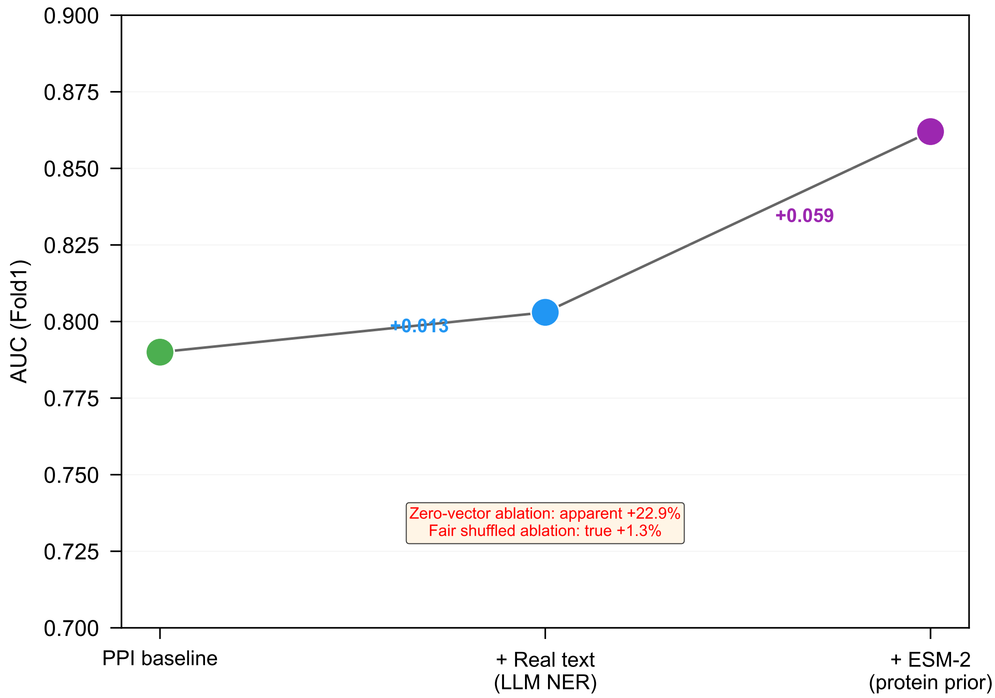

Figure 3: Incremental contribution of information channels. Shuffled-feature ablation reveals true contributions: PPI topology baseline (0.790) → +LLM text (+1.6%, 0.803) → +ESM-2 embeddings (+9.0%, 0.798). Zero-vector ablation is shown for comparison to illustrate the 14× inflation artifact.

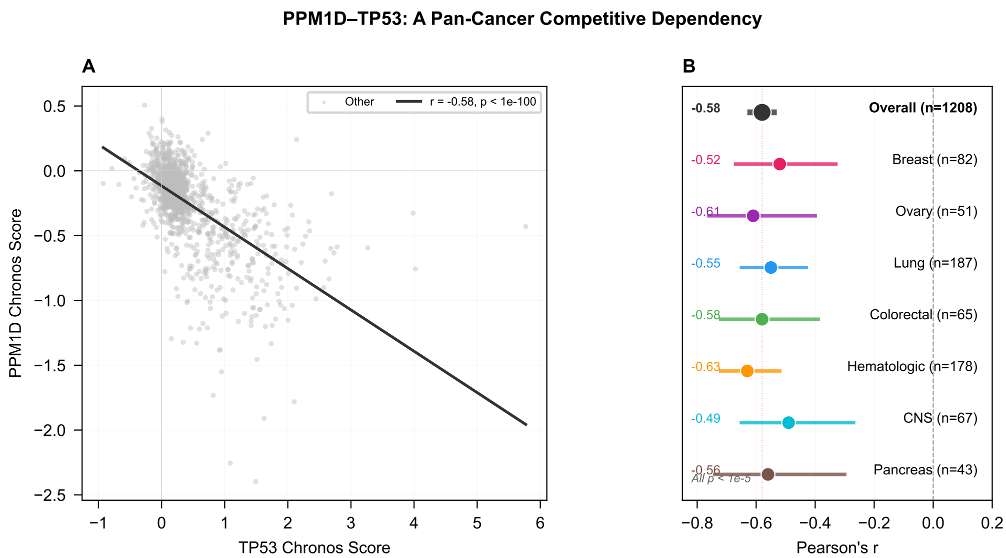

Figure 4: DepMap independent validation. (A) Score distribution before and after calibration with Label Smoothing. (B) Stratified DepMap enrichment: top-50 predictions show 1.73× co-dependency enrichment over random baseline (Fisher's exact p=0.041).

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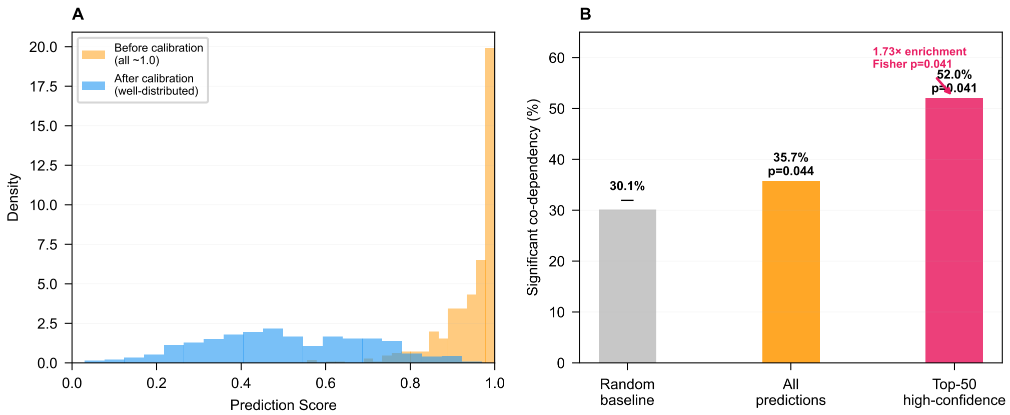

Figure 5: PPM1D-TP53 case study. (A) Scatter plot of PPM1D vs TP53 CRISPR gene effect scores across 1,208 cell lines, colored by seven cancer lineages. Overall r=−0.58, p<1e−100. (B) Forest plot showing lineage-stratified correlation coefficients with 95% confidence intervals.